DATABASE PROJECT PROGRESS REPORT

Supply Chain and Warehouse Logistics Network

Phase 1 Conceptual Design

This report presents the conceptual design of a database that tracks inventory across warehouses in different geographical locations, supplier orders, shipments, and delivery events. The model addresses two central requirements: multi-level product structures and restock alerts when inventory falls below its reorder level.

The Phase 1 result defines the system scope, 11 functional requirements, 15 business rules, 13 entities, and 17 relationships with explicit cardinality and participation. It also provides a complete EER model and conceptual logic for recursive product components and inventory alerts.

Information	Details
Course	Database Systems
Topic	6 Supply Chain and Warehouse Logistics Network
Stage	Phase 1 Conceptual Design
Submission	Progress report and EER model

Completed Work

- Defined the problem, users, scope, and assumptions of the system.
- Modelled warehouses, products, suppliers, supplier orders, shipments, tracking events, and inventory.
- Represented multi-level product structures through the recursive ProductComponent entity.
- Specified when a restock alert must be created, retained, closed, and opened again.
- Checked completeness with acceptance cases at the end of the report.

I. Problem Context and Scope

1.1 Problem Statement

The organization operates warehouses in different cities or countries. A product may be stored in several warehouses, a supplier may provide several products, and one supplier order may be delivered through several shipments. Fragmented records make it difficult to confirm the stock held at each location, compare ordered and delivered quantities, track shipments, and identify products that require replenishment.

Some finished products are assembled from components that may contain their own subcomponents. A flat product list cannot represent this structure or support material

requirement calculations. The database must support an unlimited number of component levels while preventing a product from containing itself directly or indirectly.

1.2 Objectives

- Manage warehouses, products, suppliers, and supplier product details in one system.
- Maintain one inventory balance for every warehouse and product combination.
- Track supplier orders, order items, partial shipments, and shipment events over time.
- Represent multi-level product structures with the quantity required at each level.
- Record every stock movement and issue an alert when inventory is below its reorder level.
- Provide a clear EER model that can be mapped to a relational design in Phase 2.

1.3 Scope and Assumptions

Category	Details
In scope	Geographical warehouses; products and suppliers; current supplier prices; supplier orders; partial shipments; tracking events; inventory; stock movements; multi-level product structures; restock alerts.
Outside Phase 1	Customer sales; route optimization; production batches; serial numbers; returns; warehouse transfers; unit conversion; BOM version history; payments and accounting.
Order assumptions	Each order has one supplier and one destination warehouse. A product appears at most once in the same order.
Data assumptions	Each product uses one base unit; prices use one common currency; event times use the same standard time zone.

II. Entities, Attributes, Relationships and Definitions

This section focuses on the definitions required for the Phase 1 conceptual model. Each entity is explained in normal prose, while the attribute, relationship, and function details are kept in tables for easier reading and checking.

2.1 System Definition

The Supply Chain and Warehouse Logistics Network Database manages products across multiple warehouses, suppliers, supplier orders, shipments, inventory transactions, multi-level product structures, and restock alerts. It provides a consistent view of what products exist, where they are stored, which suppliers offer them, what has been ordered, how orders are delivered, how inventory changes, and when stock falls below its reorder level.

2.2 User / Actor Definitions

2.2.1 Purchasing Staff: Purchasing Staff are employees of the organization assigned to the purchasing function within its supply chain operations.

2.2.2 Warehouse Staff: Warehouse Staff are employees of the organization assigned to warehouse operations at one or more warehouse locations.

2.2.3 Logistics Staff: Logistics Staff are employees of the organization assigned to the logistics function within its supply chain operations.

2.2.4 Inventory Manager: The Inventory Manager is the person assigned managerial responsibility for inventory control and replenishment planning within the organization's warehouse operations.

2.3 Users and Main Workflow

2.3.1 User Roles and Responsibilities

User Role	Main Responsibility
Purchasing Staff	Responsible for supplier-related purchasing activities, including maintaining supplier offerings and prices, creating supplier orders, and monitoring order status.
Warehouse Staff	Responsible for warehouse-side operations, including reviewing inventory, confirming physical receipts of goods, and recording stock issues and adjustments.
Logistics Staff	Responsible for shipment operations, including creating shipments, allocating order quantities, dispatching shipments, and recording tracking events.
Inventory Manager	Responsible for reviewing inventory conditions, evaluating replenishment needs, and making replenishment decisions based on stock levels and restock alerts.

2.3.2 Main Business Workflow

The main business workflow consists of the following stages:

Step 1 – Supplier Order Creation

Purchasing Staff selects a supplier and a destination warehouse, then creates a supplier order containing one or more product items, their quantities, and the agreed prices.

Step 2 – Shipment Preparation and Allocation

Logistics Staff creates shipments against the supplier order and allocates order-item quantities to each shipment. The total allocated quantity must not exceed the ordered quantity.

Step 3 – Shipment Dispatch and Tracking

Logistics Staff dispatches shipments and records tracking events to maintain shipment status and location information throughout the delivery process.

Step 4 – Goods Receipt and Inventory Update

Warehouse Staff confirms the physical receipt of delivered goods. The system records the receipt as a stock movement and updates the corresponding inventory quantity.

Step 5 – Inventory Monitoring and Restock Alerts

The system evaluates inventory quantities against their reorder levels. When the quantity falls below the reorder level, the system creates a restock alert if no unresolved alert already exists. When the quantity recovers to the reorder level or above, the system resolves the open alert.

Step 6 – Replenishment Decision

The Inventory Manager reviews inventory levels and restock alerts to determine whether replenishment is necessary. If additional goods are required, the purchasing process can begin again.

2.4 Entity Definitions

2.4.1 Warehouse

Definition: Represents a physical storage location. It stores warehouse identification and location/capacity information.
Entity type: Strong (independent) entity.
Identifier: WarehouseID
Relationship meaning: A Warehouse may have 0..N Inventory records and may be the destination of 0..N SupplierOrders.

2.4.2 Product

Definition: Represents a stocked, purchased, or assembled item.
Entity type: Strong (independent) entity.
Identifier: ProductID
Relationship meaning: A Product may have 0..N Inventory and SupplierProduct records, may appear in many SupplierOrderItems, and may participate as parent or component in ProductComponent.

2.4.3 Inventory

Definition: Represents the current stock balance of one Product at one Warehouse.
Entity type: Associative entity depending on Warehouse and Product.
Identifier: InventoryID
Relationship meaning: Each record belongs to exactly one Warehouse and one Product; the WarehouseID + ProductID pair is unique.

2.4.4 Supplier

Definition: Represents an organization or business entity that supplies products to the organization.
Entity type: Strong (independent) entity.
Identifier: SupplierID
Relationship meaning: A Supplier may offer many Products through SupplierProduct and may be referenced by many SupplierOrders.

2.4.5 SupplierProduct

Definition: Represents a supplier's offering of a product, including current price and lead time.
Entity type: Weak/Associative entity depending on Supplier and Product.
Identifier: SupplierID + ProductID
Relationship meaning: Resolves the many-to-many Supplier–Product relationship.

2.4.6 SupplierOrder

Definition: Represents a purchase order placed with a supplier.
Entity type: Strong entity (transaction header).
Identifier: OrderID
Relationship meaning: Each order has exactly one Supplier and one destination Warehouse; a submitted order must contain at least one item.

2.4.7 SupplierOrderItem

Definition: Represents one product line within a supplier order.
Entity type: Associative entity depending on SupplierOrder.

Identifier: OrderItemID
Relationship meaning: Each item belongs to exactly one order and one Product; a Product cannot repeat in the same order; UnitPrice is the historical agreed price; the SupplierOrderID + ProductID pair is unique.

2.4.8 Shipment

Definition: Represents a physical delivery made against a supplier order.
Entity type: Strong entity (transaction header).
Identifier: ShipmentID
Relationship meaning: A SupplierOrder may have 0..N Shipments; each Shipment belongs to one order and inherits its destination Warehouse.

2.4.9 ShipmentItem

Definition: Represents the quantity of an order line allocated to a shipment.
Entity type: Associative entity depending on Shipment and SupplierOrderItem.
Identifier: ShipmentItemID
Relationship meaning: Supports partial delivery; total non-cancelled allocations cannot exceed the ordered quantity.

2.4.10 TrackingEvent

Definition: Represents a shipment tracking/status event.
Entity type: Associative entity depending on Shipment.
Identifier: TrackingEventID
Relationship meaning: Each event belongs to one Shipment; a Shipment may have 0..N events.

2.4.11 ProductComponent

Definition: Represents a direct parent–component relationship in a multi-level product structure.
Entity type: Recursive associative entity Product and .
Identifier: ParentProductID + ComponentProductID
Relationship meaning: The same Product entity appears in parent and component roles; direct or indirect cycles are not allowed.

2.4.12 StockMovement

Definition: Represents a successful inventory transaction and preserves stock-change history.
Entity type: Associative entity depending on Inventory; optionally ShipmentItem.
Identifier: MovementID
Relationship meaning: Receipt movements may reference a ShipmentItem; other movement types do not.

2.4.13 RestockAlert

Definition: Represents one low-stock alert period.
Entity type: Associative entity depending on Inventory.
Identifier: AlertID
Relationship meaning: An Inventory record may have at most one unresolved alert; a later low-stock period creates a new alert after the previous one is resolved.

2.5 Attribute Definitions and Data Dictionary

2.5.1 Warehouse

Attribute	Data Type	Key	Null	Description / Constraints
WarehouseID	INT	PK	NOT NULL	Unique identifier of a warehouse.
WarehouseName	VARCHAR(100)	—	NOT NULL	Name of the warehouse.
Address	VARCHAR(200)	—	NOT NULL	Physical address of the warehouse.
City	VARCHAR(100)	—	NOT NULL	City where the warehouse is located.
Country	VARCHAR(100)	—	NOT NULL	Country where the warehouse is located.
CapacityM3	DECIMAL(12,2)	—	NOT NULL	Maximum storage capacity in cubic meters. Must not be negative.

2.5.2 Product

Attribute	Data Type	Key	Null	Description / Constraints
ProductID	INT	PK	NOT NULL	Unique identifier of a product.
ProductName	VARCHAR(150)	—	NOT NULL	Name used to identify the product.
Description	TEXT	—	NULL	Additional descriptive information about the product.
Unit	VARCHAR(20)	—	NOT NULL	Base unit used to measure the product quantity.
ProductType	VARCHAR(50)	—	NULL	Type or category of the product. Controlled list is finalized in Phase 2.

2.5.3 Inventory

Attribute	Data Type	Key	Null	Description / Constraints
InventoryID	INT	PK	NOT NULL	Unique identifier of an inventory record.
WarehouseID	INT	FK	NOT NULL	References Warehouse. Each inventory belongs to exactly one warehouse.
ProductID	INT	FK	NOT NULL	References Product. Each inventory belongs to exactly one product.
Quantity	DECIMAL(12,2)	—	NOT NULL	Current quantity available. Cannot be negative.
ReorderLevel	DECIMAL(12,2)	—	NOT NULL	Minimum inventory threshold. Cannot be negative.
LastUpdated	DATETIME	—	NOT NULL	Date and time when the inventory balance was most recently changed.

2.5.4 Supplier

Attribute	Data Type	Key	Null	Description / Constraints
SupplierID	INT	PK	NOT NULL	Unique identifier of a supplier.
SupplierName	VARCHAR(150)	—	NOT NULL	Name of the supplier.

ContactName	VARCHAR(100)	—	NULL	Name of the main contact person.
Phone	VARCHAR(30)	—	NULL	Supplier's contact telephone number.
Email	VARCHAR(150)	—	NULL	Supplier's contact email address.
Address	VARCHAR(200)	—	NULL	Supplier's business address.

2.5.5 SupplierProduct

Attribute	Data Type	Key	Null	Description / Constraints
SupplierID	INT	PK, FK	NOT NULL	References Supplier and forms part of the composite primary key.
ProductID	INT	PK, FK	NOT NULL	References Product and forms part of the composite primary key.
UnitPrice	DECIMAL(12,2)	—	NOT NULL	Current price offered by the supplier.
LeadTimeDays	INT	—	NOT NULL	Expected number of days required for delivery.

2.5.6 SupplierOrder

Attribute	Data Type	Key	Null	Description / Constraints
OrderID	INT	PK	NOT NULL	Unique identifier of a supplier order.
OrderDate	DATE	—	NOT NULL	Date on which the supplier order is created.
ExpectedDeliveryDate	DATE	—	NULL	Expected delivery date at the destination warehouse.
Status	VARCHAR(20)	—	NOT NULL	Current lifecycle state of the supplier order.
SupplierID	INT	FK	NOT NULL	References Supplier. Each order refers to exactly one supplier.
WarehouseID	INT	FK	NOT NULL	References Warehouse. Each order has exactly one destination warehouse.

2.5.7 SupplierOrderItem

Attribute	Data Type	Key	Null	Description / Constraints
OrderItemID	INT	PK	NOT NULL	Unique identifier of an individual order line.
OrderID	INT	FK	NOT NULL	References SupplierOrder.
ProductID	INT	FK	NOT NULL	References Product.
Quantity	DECIMAL(12,2)	—	NOT NULL	Quantity of the product requested in the order line.
UnitPrice	DECIMAL(12,2)	—	NOT NULL	Historical price agreed when the order was created.

2.5.8 Shipment

Attribute	Data Type	Key	Null	Description / Constraints
ShipmentID	INT	PK	NOT NULL	Unique identifier of a shipment.
OrderID	INT	FK	NOT	References SupplierOrder. One order

			NULL	may have multiple shipments.
TrackingNumber	VARCHAR(50)	—	NULL	External tracking reference assigned to the shipment.
ShipmentDate	DATE	—	NULL	Date when the shipment is dispatched.
EstimatedArrival	DATETIME	—	NULL	Current expected arrival date and time.
ActualArrival	DATETIME	—	NULL	Confirmed arrival date and time. Empty until arrival is confirmed.
Status	VARCHAR(20)	—	NOT NULL	Current logistics state of the shipment.

2.5.9 ShipmentItem

Attribute	Data Type	Key	Null	Description / Constraints
ShipmentItemID	INT	PK	NOT NULL	Unique identifier of a shipment line.
ShipmentID	INT	FK	NOT NULL	References Shipment.
OrderItemID	INT	FK	NOT NULL	References SupplierOrderItem. Must belong to the same supplier order as the shipment.
Quantity	DECIMAL(12,2)	—	NOT NULL	Quantity allocated to this shipment. Must be positive. Total allocation cannot exceed the originally ordered quantity.

2.5.10 TrackingEvent

Attribute	Data Type	Key	Null	Description / Constraints
TrackingEventID	INT	PK	NOT NULL	Unique identifier of a tracking event.
ShipmentID	INT	FK	NOT NULL	References Shipment.
EventTime	DATETIME	—	NOT NULL	Date and time when the tracking event occurs.
Location	VARCHAR(150)	—	NULL	Geographical location where the event is recorded.
Status	VARCHAR(30)	—	NOT NULL	Shipment status reported by the tracking event.
Description	TEXT	—	NULL	Additional information about the event or handling activity.

2.5.11 ProductComponent

Attribute	Data Type	Key	Null	Description / Constraints
ParentProductID	INT	PK, FK	NOT NULL	References Product as the parent/assembled product.
ComponentProductID	INT	PK, FK	NOT NULL	References Product as the component product.
QuantityRequired	DECIMAL(12,2)	—	NOT NULL	Number of component units required to produce one parent product. Must be

				greater than zero.

2.5.12 StockMovement

Attribute	Data Type	Key	Null	Description / Constraints
MovementID	INT	PK	NOT NULL	Unique identifier of an inventory transaction.
InventoryID	INT	FK	NOT NULL	References Inventory.
ShipmentItemID	INT	FK	NULL	Optional reference to ShipmentItem. Used only for Receipt movements.
MovementTime	DATETIME	—	NOT NULL	Date and time when the inventory transaction is recorded.
MovementType	VARCHAR(20)	—	NOT NULL	Category of inventory change, e.g. Opening, Receipt, Issue, Adjustment.
QuantityDelta	DECIMAL(12,2)	—	NOT NULL	Signed inventory change. Positive increases inventory; negative decreases inventory.

2.5.13 RestockAlert

Attribute	Data Type	Key	Null	Description / Constraints
AlertID	INT	PK	NOT NULL	Unique identifier of a low-stock alert period.
InventoryID	INT	FK	NOT NULL	References Inventory.
AlertDate	DATETIME	—	NOT NULL	Date and time when the low-stock condition is first detected.
QuantityAtAlert	DECIMAL(12,2)	—	NOT NULL	Inventory quantity captured when the alert is created. Historical snapshot.
ReorderLevelAtAlert	DECIMAL(12,2)	—	NOT NULL	Reorder level captured when the alert is created. Historical snapshot.
ResolvedAt	DATETIME	—	NULL	Date and time when the alert is resolved. NULL means unresolved.
Message	VARCHAR(200)	—	NULL	Human-readable explanation of the low-stock condition.

III. Functional Requirements

ID	Function	Requirement
FR01	Warehouse management	The system shall allow authorized users to create, view, and update warehouse information, including warehouseID, name, address, city, country, and capacity of every warehouses.
FR02	Product management	The system shall allow authorized users to create, view, and update product information, including ProductID, product name, description, base unit, and product type, according to the defined Product

		entity. Each product shall be uniquely identifiable.
FR03	Supplier and SupplierProduct management	The system shall allow authorized users to maintain supplier information and manage the products offered by each supplier. For each supplier-product offering, the system shall store the current unit price and lead time. The system shall support the many-to-many relationship between suppliers and products.
FR04	Supplier orders management	The system shall allow authorized users to create and manage supplier orders, assign each order to exactly one supplier and one destination warehouse, and add or remove product items with their ordered quantities and agreed unit prices.
FR05	Shipment Management and Partial deliveries	The system shall allow authorized users to create shipments for supplier orders and allocate order-item quantities to shipment items. A single order item may be delivered through multiple shipments. The total quantity allocated across non-cancelled shipments shall not exceed the ordered quantity.
FR06	Shipment tracking	The system shall maintain shipment tracking information, including the tracking number, dispatch date, expected arrival date, actual arrival date, and current status. It shall allow authorized users to record and retrieve chronological tracking events associated with each shipment, including relevant status and location information.
FR07	Inventory management	The system shall maintain an inventory record for each warehouse-product combination, including the current quantity and reorder level. It shall allow authorized users to view inventory quantities by warehouse and product and support inventory monitoring across multiple warehouses.
FR08	Stock movements	The system shall record inventory changes resulting from opening balances, goods receipts, stock issues, and stock adjustments. Each successful inventory-changing transaction shall create exactly one corresponding StockMovement record. When goods are received, the system shall update the relevant inventory quantity and associate the receipt movement with the corresponding ShipmentItem.
FR09	Product structure management	The system shall allow authorized users to define and maintain product-component relationships through the ProductComponent entity. Each relationship shall identify a parent product, a component product, and the required quantity of that component.

FR10	BOM cycle prevention	The system shall validate product-component relationships before accepting them and reject any relationship that would introduce a direct or indirect cycle into the Bill of Materials (BOM).
FR11	Restock alerts management	The system shall automatically create a restock alert when an inventory quantity falls below its reorder level and no unresolved alert exists for that inventory record. It shall retain the unresolved alert while the low-stock condition persists and resolve the alert when the inventory quantity reaches or exceeds the reorder level. If the inventory later falls below the reorder level again, the system shall allow a new alert to be created for the new low-stock period.

IV. Business Rules

4.1. Business Constraints

ID	Business Constraint
BR01	Each Inventory record belongs to exactly one Warehouse and one Product. A warehouse or product may have zero or many inventory records. WarehouseID and ProductID must be unique as a pair.
BR02	Supplier and Product have a many-to-many relationship through SupplierProduct. Each supplier product record stores the current UnitPrice and LeadTimeDays.
BR03	Each SupplierOrder refers to exactly one Supplier and one destination Warehouse. A supplier or warehouse may have zero or many orders.
BR04	A draft order may have no items, but a submitted order must have at least one. Each SupplierOrderItem belongs to exactly one order and one product, and a product cannot repeat in the same order.
BR05	The order supplier must have a SupplierProduct record for every ordered product. UnitPrice on SupplierOrderItem is the agreed historical price and does not change with the current supplier price.
BR06	An order may have zero or many Shipments. Each Shipment belongs to exactly one order and inherits the destination warehouse from that order.
BR07	A draft shipment may have no items, but a dispatched shipment must have at least one ShipmentItem. Each ShipmentItem refers to a SupplierOrderItem from the same order as the shipment.
BR08	The total quantity allocated to shipment items belonging to non-cancelled shipments shall not exceed the ordered quantity of the corresponding supplier order item.
BR09	Each TrackingEvent belongs to exactly one Shipment. A shipment may have zero or many events. TrackingEventID distinguishes events that occur at the same time.
BR10	Product contains Product is a recursive many-to-many relationship. ProductComponent stores the parent product, component product, and a QuantityRequired value greater than zero.

ID	Business Constraint
BR11	A product must not contain itself directly or indirectly. A product without components is a leaf item, while a component may still contain lower-level components.
BR12	Inventory.Quantity and ReorderLevel cannot be negative. A stock issue is rejected if it would make Quantity negative. Each successful change creates exactly one StockMovement.
BR13	A Receipt movement refers to exactly one ShipmentItem, and each ShipmentItem may be received at most once. Other movement types do not refer to a shipment item.
BR14	Low stock occurs when Quantity is less than ReorderLevel. Each Inventory record may have at most one unresolved RestockAlert. Repeated low-stock updates must not create duplicate alerts.
BR15	When Quantity is greater than or equal to ReorderLevel, the open alert is resolved. If Quantity later falls below the threshold, a new alert period begins. A change to ReorderLevel must also be evaluated.

4.2 Relationship Definitions, Cardinality and Participation

ID	Relationship	Cardinality	Participation / Meaning
R01	Warehouse – Inventory	1 : 0..N	A Warehouse may have zero or many Inventory records. Each Inventory record belongs to exactly one Warehouse.
R02	Product – Inventory	1 : 0..N	A Product may have zero or many Inventory records. Each Inventory record refers to exactly one Product.
R03	Supplier – SupplierProduct	1 : 0..N	A Supplier may have zero or many SupplierProduct records. Each SupplierProduct record belongs to exactly one Supplier
R04	Product – SupplierProduct	1 : 0..N	A Product may appear in zero or many SupplierProduct records. Each SupplierProduct record refers to exactly one Product.
R05	Supplier – SupplierOrder	1 : 0..N	A Supplier may be associated with zero or many SupplierOrders. Each SupplierOrder refers to exactly one Supplier.
R06	Warehouse – SupplierOrder	1 : 0..N	A Warehouse may be the destination of zero or many SupplierOrders. Each SupplierOrder has exactly one destination Warehouse.
R07	SupplierOrder – SupplierOrderItem	1 : 0..N	A SupplierOrder may contain zero or many SupplierOrderItems. Each SupplierOrderItem belongs to exactly one SupplierOrder. A Draft order may be empty, but a Submitted order must contain at least one item.
R08	Product – SupplierOrderItem	1 : 0..N	A Product may appear in zero or many SupplierOrderItems. Each SupplierOrderItem refers to exactly one Product.
R09	SupplierOrder – Shipment	1 : 0..N	A SupplierOrder may have zero or many Shipments. Each Shipment belongs to exactly one SupplierOrder.
R10	Shipment – ShipmentItem	1 : 0..N	A Shipment may contain zero or many ShipmentItems. Each ShipmentItem belongs to exactly one Shipment. A Draft shipment may be empty, but a Dispatched shipment must contain at least one item.
R11	SupplierOrderItem - ShipmentItem	1 : 0..N	A SupplierOrderItem may be allocated to zero or many ShipmentItems. Each ShipmentItem refers to exactly one SupplierOrderItem.
R12	Shipment –	1 : 0..N	A Shipment may have zero or many TrackingEvents.

	TrackingEvent		Each TrackingEvent belongs to exactly one Shipment.
R13	Parent Product – ProductComponent	1 : 0..N	A Product may act as the parent in zero or many ProductComponent records. Each ProductComponent record has exactly one Parent Product.
R14	Component Product – ProductComponent	1 : 0..N	A Product may act as the component in zero or many ProductComponent records. Each ProductComponent record identifies exactly one Component Product.
R15	Inventory – StockMovement	1 : 0..N	An Inventory record may have zero or many StockMovements. Each StockMovement belongs to exactly one Inventory record.
R16	Inventory – RestockAlert	1 : 0..N	An Inventory record may have zero or many RestockAlerts over time. Each RestockAlert belongs to exactly one Inventory record. However, no more than one alert for the same Inventory record may remain unresolved at a time.
R17	ShipmentItem - StockMovement	0..1 : 0..1	A ShipmentItem may be linked to zero or one StockMovement, and a StockMovement may reference zero or one ShipmentItem. Only a Receipt StockMovement may reference a ShipmentItem, and each ShipmentItem may be received at most once.

4.3 Key Business Constraints

Constraint	Definition	Rule
Inventory uniqueness	One Warehouse–Product pair has at most one Inventory record.	BR01
Supplier-product validity	The order supplier must offer every ordered product.	BR05
Order uniqueness	A Product may appear at most once in a SupplierOrder. The SupplierOrderID–ProductID pair in SupplierOrderItem must therefore be unique.	BR04
Partial shipment limit	For each SupplierOrderItem, the total quantity allocated to ShipmentItems in non-cancelled Shipments must not exceed the ordered quantity.	BR08
Shipment-order consistency	Each ShipmentItem must reference a SupplierOrderItem belonging to the same SupplierOrder as its associated Shipment.	BR07
Inventory non-negativity	Quantity and ReorderLevel cannot be negative; invalid issues are rejected.	BR12
Receipt uniqueness	Each Receipt StockMovement must reference exactly one ShipmentItem, and each ShipmentItem may be received at most once. Other StockMovement types must not reference a ShipmentItem.	BR13
Recursive structure validity	No direct/indirect Product cycle; QuantityRequired > 0.	BR10–BR11
Restock alert uniqueness	At most one unresolved RestockAlert per Inventory.	BR14
Alert lifecycle	Alert uniqueness: Each Inventory may have at most one unresolved RestockAlert at a time.Alert lifecycle: An unresolved alert must be resolved when Quantity becomes greater than or equal to ReorderLevel. If Quantity later falls	BR14–BR15

	below ReorderLevel again, a new alert period must be created. Changes to ReorderLevel must also trigger reevaluation.	

4.4 Important Modelling Clarifications

Entity vs Actor: Warehouse, Product, Inventory, Supplier and the other listed objects are database entities. Purchasing Staff, Warehouse Staff, Logistics Staff and Inventory Manager are actors/roles.

Current state vs history: Inventory stores the current operational balance, while StockMovement stores the history of successful inventory changes and RestockAlert stores alert periods.

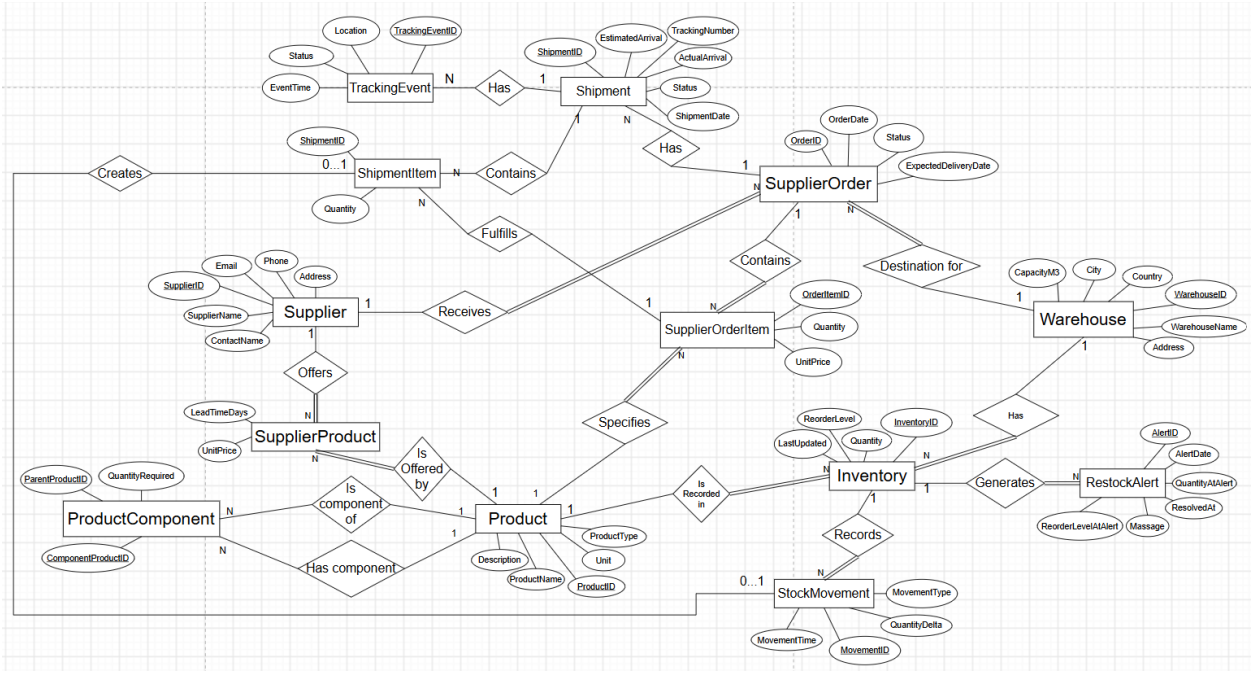
Current price vs historical price: SupplierProduct.UnitPrice is the current supplier quotation; SupplierOrderItem.UnitPrice is the agreed price captured for that order.

User action vs system action: A user initiates a valid business action; the system then performs required automatic updates and validations.

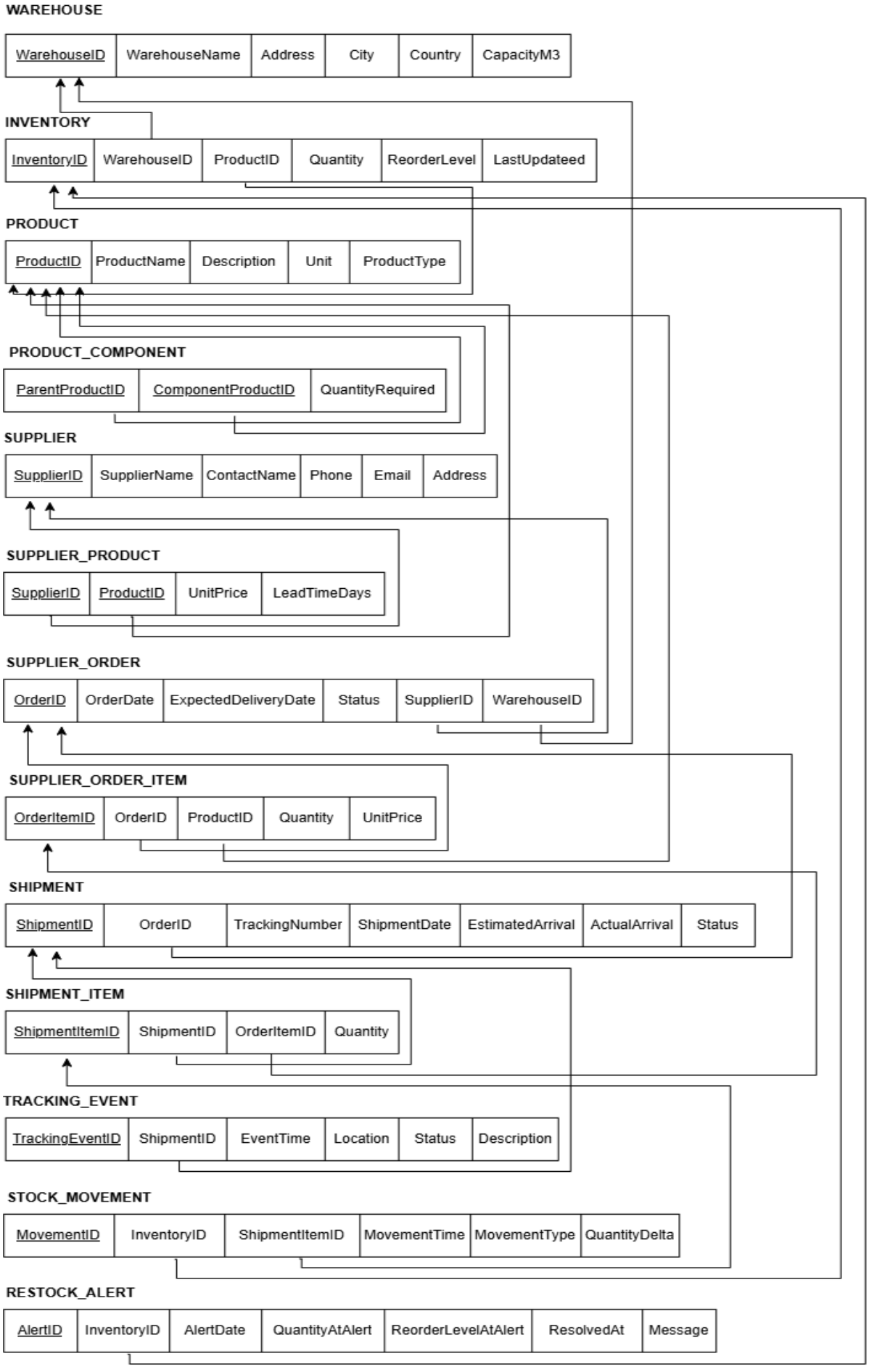
Recursive roles: ParentProductID and ComponentProductID refer to the same Product entity but represent different roles.

Phase 1 vs Phase 2: Phase 1 defines the conceptual model and business constraints. DBMS-specific SQL constraints, triggers, indexes and implementation details are deferred to Phase 2.

V. Extended Entity Relationship Model



VI. Relational Scheme


WAREHOUSE
WarehouseID
WarehouseName
Address
City
Country
CapacityM3
INVENTORY
InventoryID
WarehouseID
ProductID
Quantity
ReorderLevel
LastUpdateteed
PRODUCT
ProductID
ProductName
Description
Unit
ProductType
PRODUCT_COMPONENT
ParentProductID
ComponentProductID
QuantityRequired
SUPPLIER
SupplierID
SupplierName
ContactName
Phone
Email
Address
SUPPLIER_PRODUCT
SupplierID
ProductID
UnitPrice
LeadTimeDays
SUPPLIER_ORDER
OrderID
OrderDate
ExpectedDeliveryDate
Status
SupplierID
WarehouseID
SUPPLIER_ORDER_ITEM
OrderItemID
OrderID
ProductID
Quantity
UnitPrice
SHIPMENT
ShipmentID
OrderID
TrackingNumber
ShipmentDate
EstimatedArrival
ActualArrival
Status
SHIPMENT_ITEM
ShipmentItemID
ShipmentID
OrderItemID
Quantity
TRACKING_EVENT
TrackingEventID
ShipmentID
EventTime
Location
Status
Description
STOCK_MOVEMENT
MovementID
InventoryID
ShipmentItemID
MovementTime
MovementType
QuantityDelta
RESTOCK_ALERT
AlertID
InventoryID
AlertDate
QuantityAtAlert
ReorderLevelAtAlert
ResolvedAt
Message